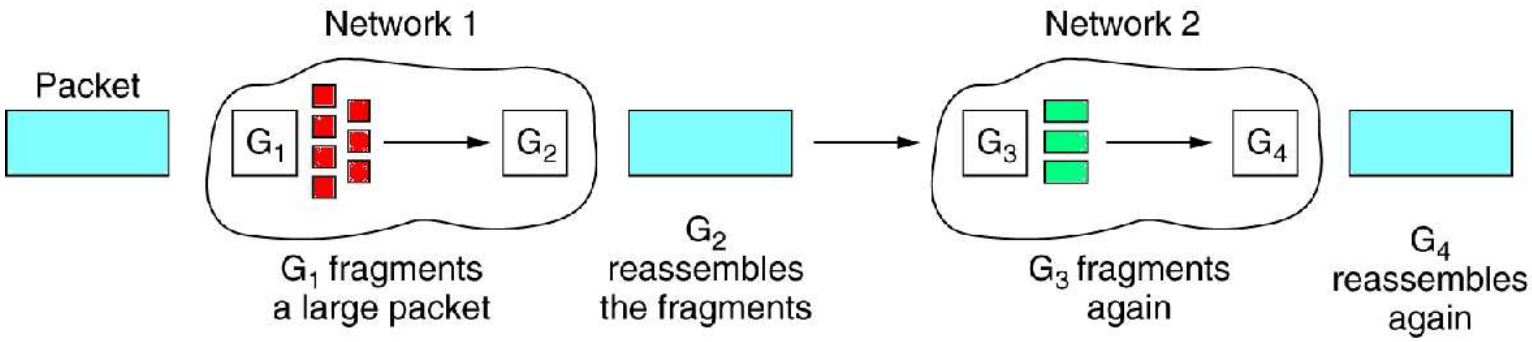
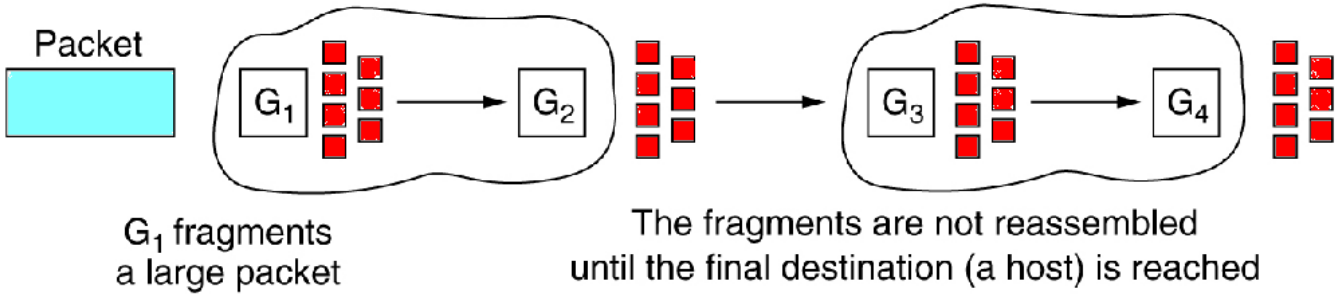
Two Types of Fragmentation



(a) Transparent Fragmentation



(b) Non-Transparent Fragmentation

What do you mean by fragmentation of packets? Distinguish between its two forms of recombination.

Fragmentation:

Each network imposes some maximum size on its packets. These limits have various causes like-

1. Hardware (e.g., the size of an Ethernet frame).
2. Operating system (e.g., all buffers are 512 bytes).
3. Protocols (e.g., the number of bits in the packet length field).
4. Compliance with some (inter)national standard.
5. Desire to reduce error-induced retransmissions to some level.
6. Desire to prevent one packet from occupying the channel too long.

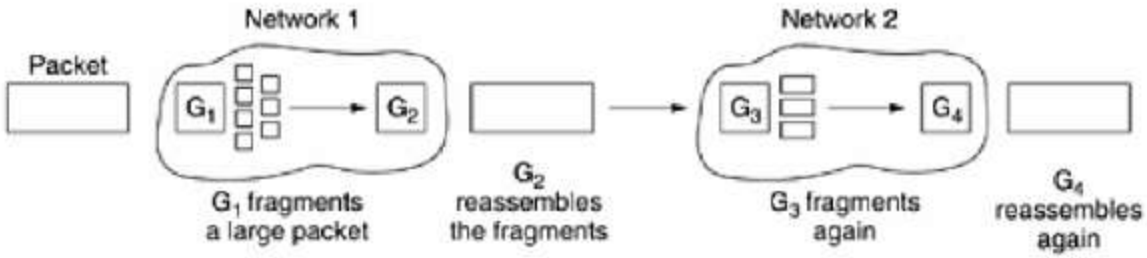
Thus, the network designers are not free to choose any maximum packet size they wish. An obvious problem appears when a large packet wants to travel through a network whose maximum packet size is too small. The only solution to the problem is to allow gateways to break up packets into fragments, sending each fragment as a separate internet packet. This solution is known as fragmentation of packets.

For recombining the packets, there are two strategies.

1. Transparent Fragmentation
2. Non-transparent Fragmentation

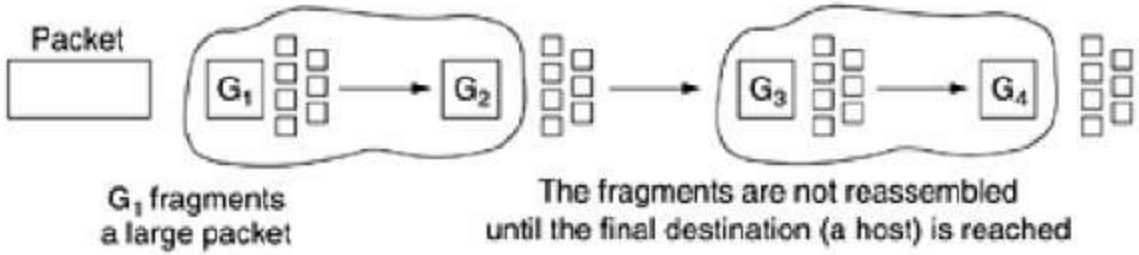
Transparent Fragmentation:

In this approach, the small-packet network has gateways that interface to other networks. When an oversized packet arrives at a gateway, the gateway breaks it up into fragments. Each fragment is addressed to the same exit gateway, where the pieces are recombined. In this way passage through the small-packet network has been made transparent. Subsequent networks are not even aware that fragmentation has occurred. ATM is an example of such fragmentation. [segmentation]



Nontransparent Fragmentation:

In this mode, once a packet has been fragmented, each fragment is treated as though it were an original packet. All fragments are passed through the exit gateway as shown below. Recombination occurs only at the destination host. IP works this way.



Discuss the various problem raised in case of fragmentation.

 # In case of Transparent Fragmentation:

- The exit gateway must know when it has received all the pieces.
- All packets must exit via the same gateway.
- The overhead required to repeatedly reassemble and refragment a large packet passing through a series of small-packet networks.

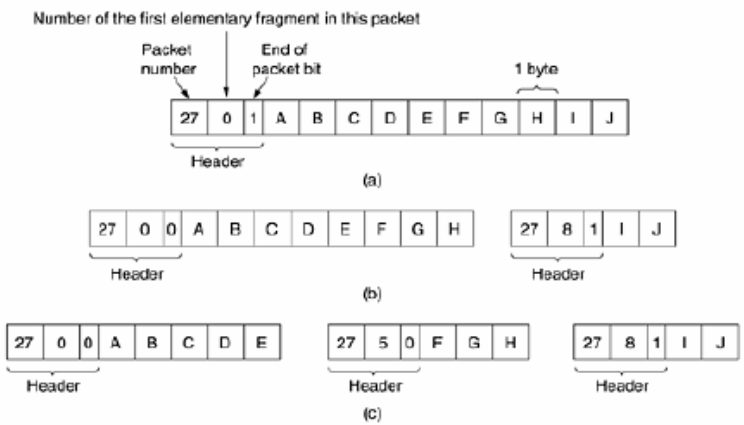
 # In case of Nontransparent Fragmentation:

- Requires every host to be able to do reassembly.
- Total overhead increases when large packets are fragmented.
- Multiple exit gateways can be used and higher performance can be achieved.[adv]

How fragments of a packet are numbered? Discuss briefly.

One way of numbering the fragments is to use a tree. If packet 0 must be split up, the pieces are called 0.0, 0.1, 0.2, etc. If these fragments themselves must be fragmented later on, the pieces are numbered 0.0.0, 0.0.1, 0.0.2, . . . , 0.1.0, 0.1.1, 0.1.2, etc. But the problem with this approach is as follows. Suppose that a 1024-bit packet is initially fragmented into four equal-sized fragments, 0.0, 0.1, 0.2, and 0.3. Fragment 0.1 is lost, but the other parts arrive at the destination. Eventually, the source times out and retransmit the original packet again. Only this time Murphy's Law strikes and the route taken passes through a network with a 512-bit limit, so two fragments are generated. When the new fragment 0.1 arrives at the destination, the receiver will think that all four pieces are now accounted for and reconstruct the packet incorrectly.

A different and better numbering system is to define an elementary fragment size small enough that the elementary fragment can pass through every network. When a packet is fragmented, all the pieces are equal to the elementary fragment size except the last one, which may be shorter. An internet packet may contain several fragments, for efficiency reasons. The internet header must provide the original packet number and the number of the (first) elementary fragment contained in the packet. As usual, there must also be a bit indicating that the last elementary fragment contained within the internet packet is the last one of the original packet. There is clearly a trade-off between the size of the elementary fragment and the number of bits in the fragment number. Because the elementary fragment size is presumed to be acceptable to every network, subsequent fragmentation of an internet packet containing several fragments causes no problem. The ultimate limit here is to have the elementary fragment be a single bit or byte, with the fragment number then being the bit or byte offset within the original packet, as shown below.



IP reassembly timer is all or nothing — Justify.

Either all fragments arrive and IP reassembles the datagram or IP discards the complete datagram. There is no mechanism for receiver to tell sender which fragments have arrived.

Even sender doesn't know about fragmentation.

A receiver can't hold fragments for long time as it occupies memory. So IP specifies a maximum time to hold fragments.

When first fragment arrives timer is started. If all fragments arrive before time expires, receiver cancels timer and reassembles datagram. Otherwise it discards those fragments.

Transparent Fragmentation

- Small-packet network transparent to other subsequent networks
- ATM networks (through special hardware) provide transparent fragmentation
- Issues:
  - Some performance penalty but requiring all fragments to go through same gateway
  - May have to repeatedly fragment and reassemble through series of small-packet networks

Non-Transparent Fragmentation

- Only reassemble at destination host
  - Each fragment becomes a separate packet
  - Thus routed independently
- Problems:
  - Every fragment must carry header until it reaches destination host
  - Fragments must be numbered so that original data stream can be reconstructed